Predicting the full-load power output of a combined cycle power plant

Machine learning regression on 9,527 hours of full-load plant data

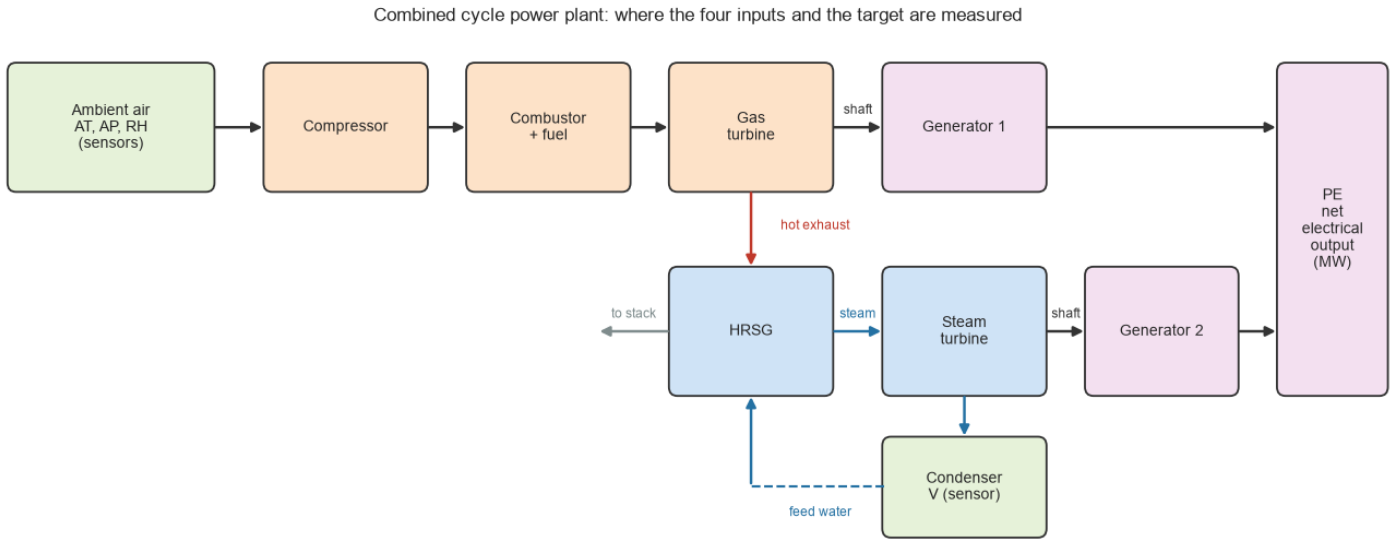
9,527
hours of full-load data

3
machine learning models

3.45 MW
test error of the chosen model

Shaikh Muhammad Zain

At full load, the weather still moves output by 76 MW



Inputs: ambient temperature AT, ambient pressure AP and relative humidity RH at the compressor inlet; exhaust vacuum V at the condenser.

Target: net electrical output PE, 420 to 496 MW.

Task: supervised regression on hourly averages.

Grid commitment

The plant promises its MW in advance. Over-predicting means a shortfall the grid must cover, usually with a penalty.

Performance monitoring

The model gives what a healthy plant should produce now. Output that stays below it points to fouling or wear.

The raw data needed four fixes before any model saw it

41

duplicate records

dropped: they would sit in both train and test sets

55

humidity readings above 100%

capped at 100%, rows kept

182

hours with V above air pressure

unit label is wrong; V kept as a back-pressure index

5

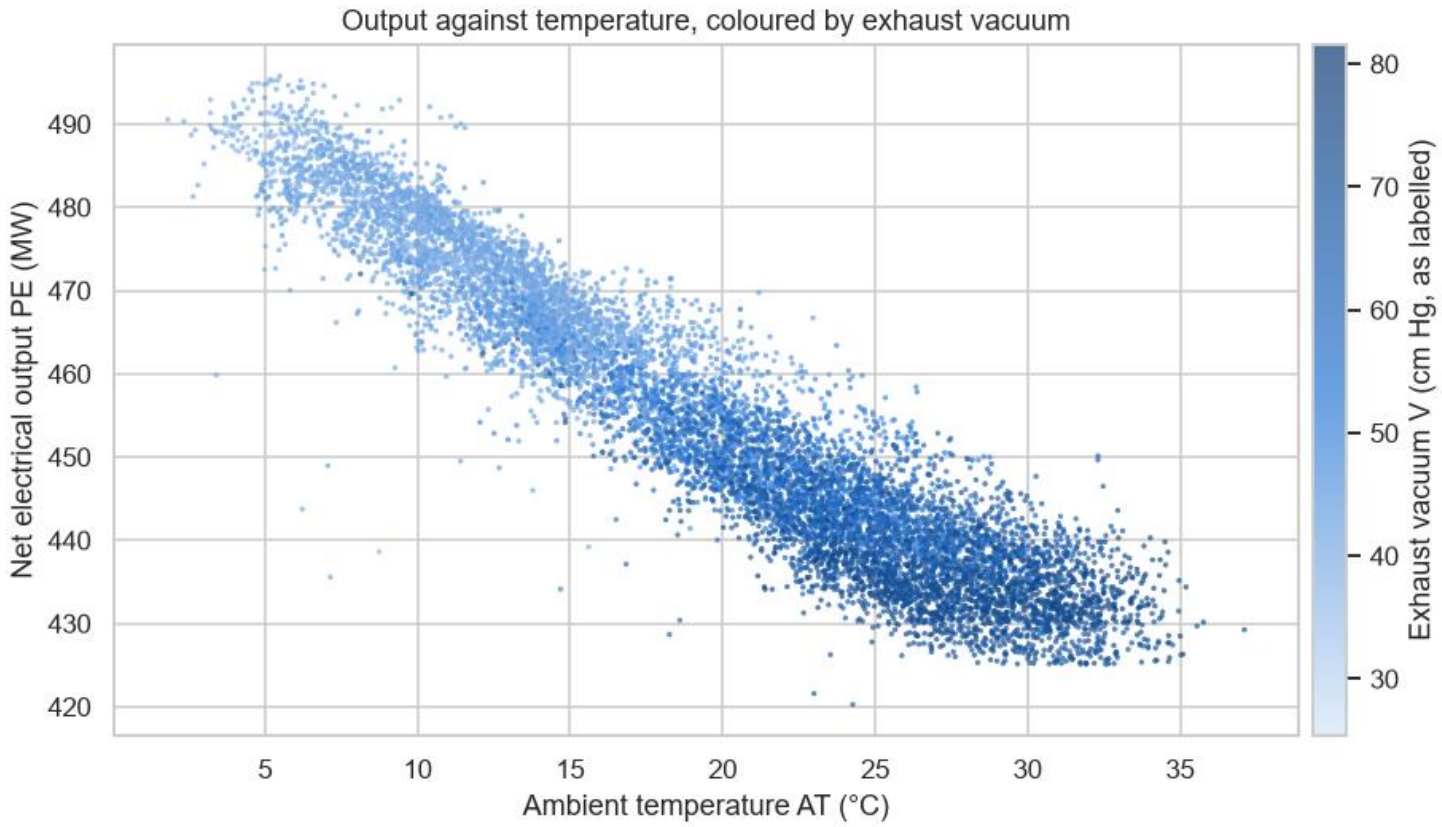
Excel sheets, identical rows

Sheet1 only; stacking would copy every hour five times

Leakage control

1. 80/20 split stratified on output: 7,621 train, 1,906 test hours
2. Scaler fitted inside the pipeline, on training folds only
3. Hyperparameters tuned by 5-fold cross-validation
4. Test set used once, after all tuning

Temperature sets the capacity: about 2.2 MW lost per °C



−2.17 MW/°C

average loss of output per degree of ambient temperature

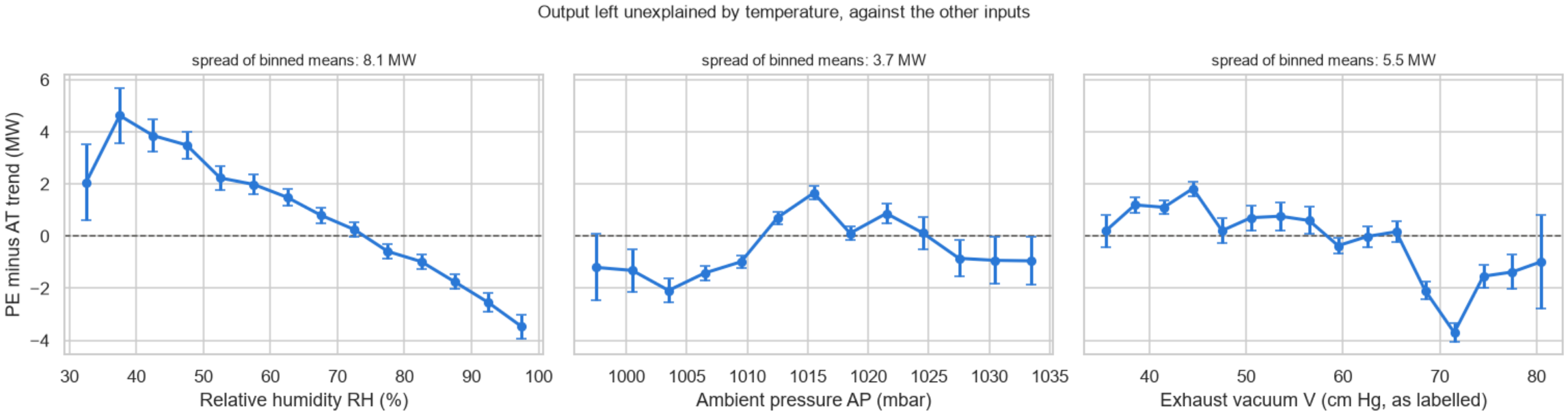
52 MW

less output in the hottest 5% of hours than in the coldest 5%

86%

of that drop is explained by the lower density of hot inlet air

Three effects only appear once temperature is held fixed



Humidity lowers output

Plain correlation +0.39, partial −0.36: humid hours are cool hours. About 8 MW from dry to saturated air.

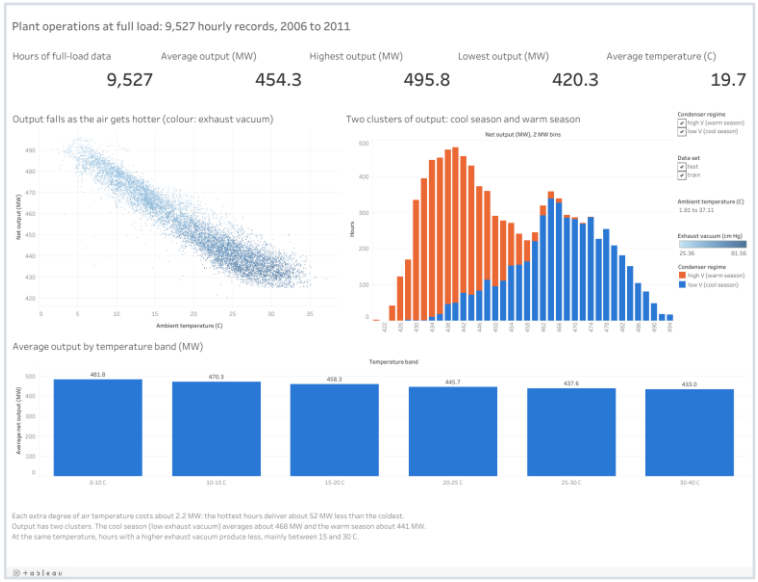
Pressure barely matters

Correlation +0.52 falls to +0.07 with the other inputs fixed. Its apparent effect is borrowed from temperature.

The condenser has a threshold

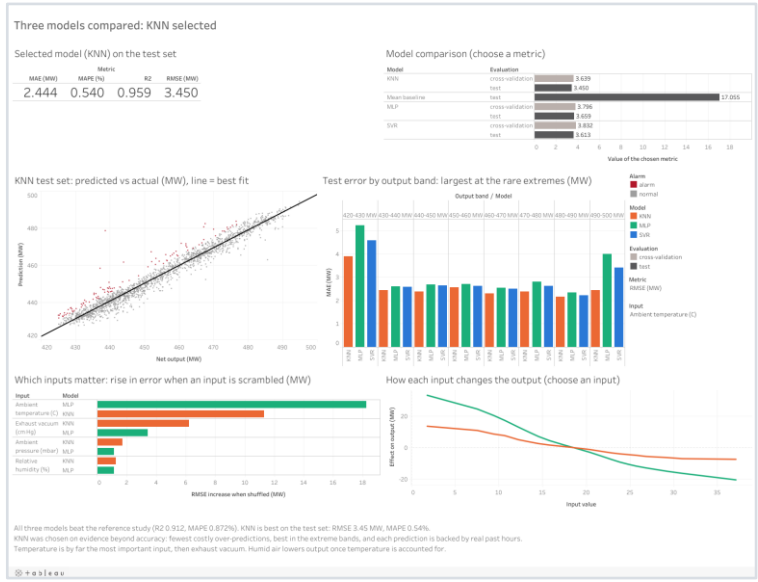
Output drops above V ≈ 66, and V matters mainly between 15 and 30 °C: an interaction with temperature.

One Tableau dashboard for data, models and decisions



Plant operations

KPIs, filters, click a temperature band to filter every chart



Model performance

metric selector, test errors by band, input effects



Decision support

adjustable alarm threshold and safety margin

Live dashboard: public.tableau.com, profile shaikh.muhammad.zain

Three nonlinear models, none used in the reference study

SVR

Support vector regression

Smooth RBF kernel follows the curved, interacting effects; errors under ε are treated as noise.

Tuned settings

C = 30, γ = 2, ε = 2 MW

Cross-validated RMSE

3.83 MW

KNN

k-nearest neighbours

Predicts from the most similar logged hours, so every prediction can be traced to real past hours.

Tuned settings

k = 6, distance-weighted, Manhattan

Cross-validated RMSE

3.64 MW

MLP

Neural network

Learns any shape from data, stays small once trained, and scales to years of sensor data.

Tuned settings

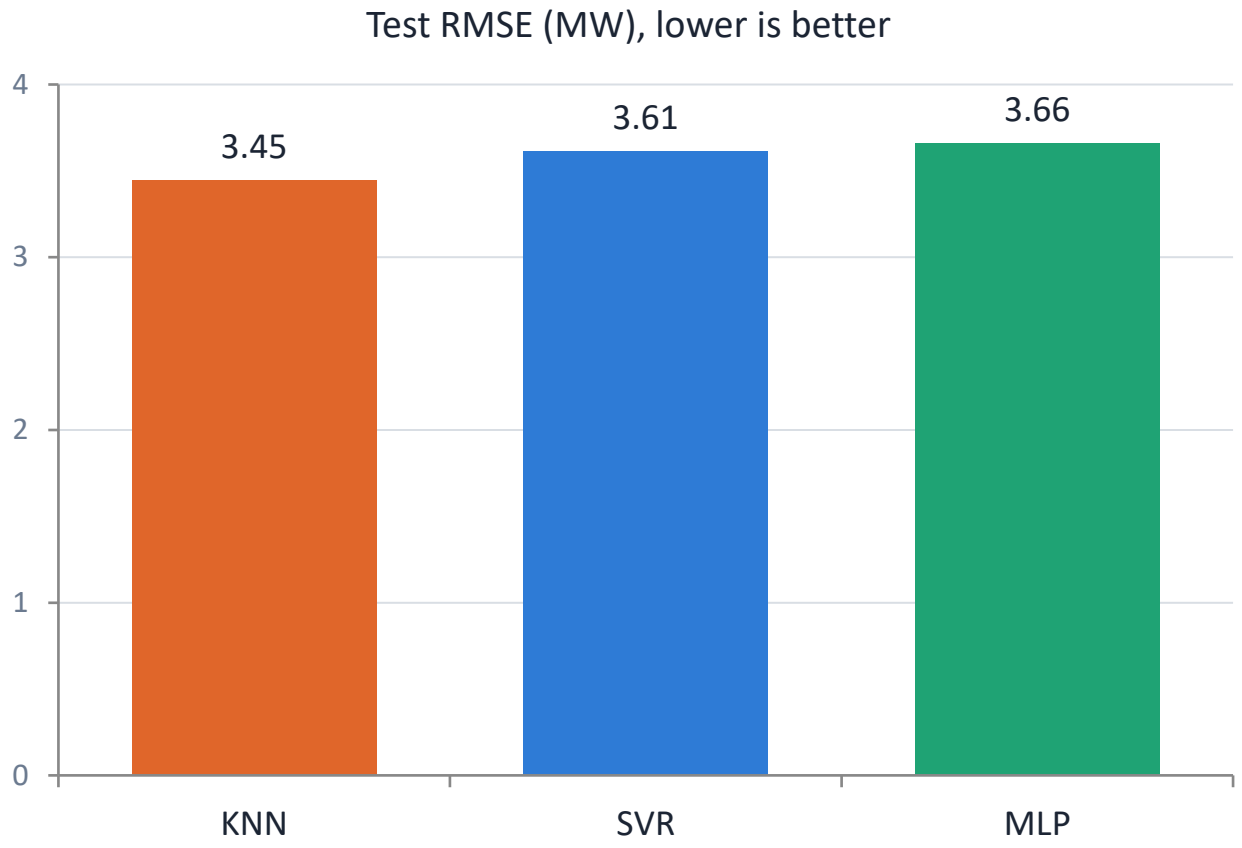
128-128-64 ReLU, early stopping

Cross-validated RMSE

3.80 MW

All three share one pipeline and one set of folds. Tuning used the 7,621 training hours only; the mean-predictor baseline scores 17.04 MW.

On 1,906 unseen hours all three beat the reference study



0.959

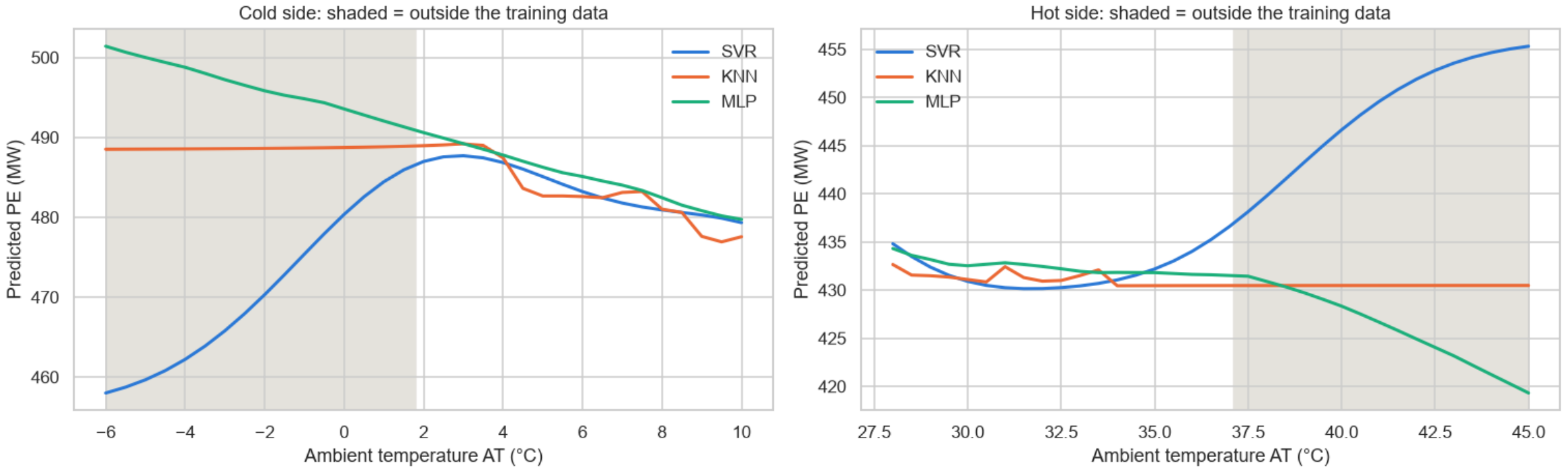
R^2 of KNN (reference 0.912)

0.540%

MAPE of KNN (reference 0.872%)

- KNN's lead is real: a paired bootstrap puts it 0.17 MW ahead of SVR and 0.21 MW ahead of MLP, with neither interval including zero.
- Test scores match cross-validation, so tuning did not overfit.
- The lead is small next to the 2.4 MW typical error, so accuracy alone did not decide the choice.

Accuracy is not enough: the SVR fails in a heatwave

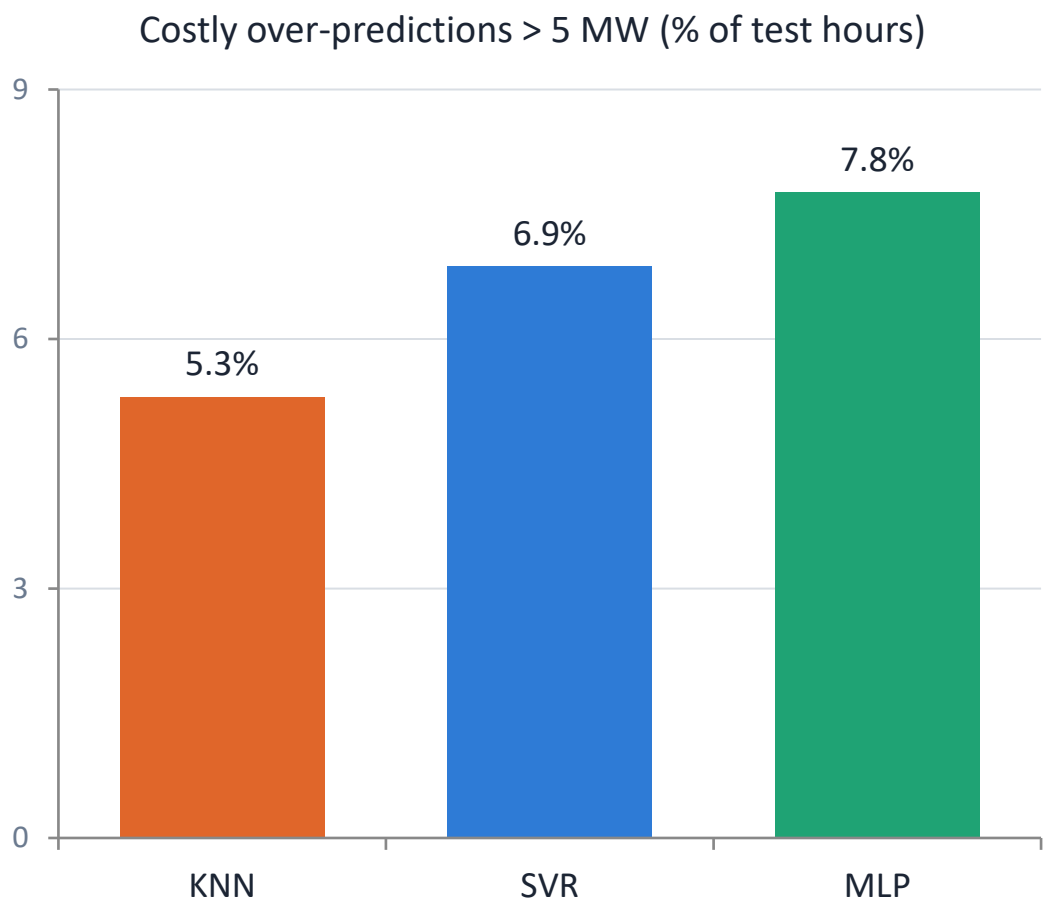


SVR turns upwards beyond 35 °C: 455 MW at 45 °C, a mild day's output. Physically wrong.

KNN stays flat at the hottest logged hours (430.5 MW). Wrong slope, safe direction.

MLP follows physics on both sides, but predicts over 500 MW in extreme cold. Unverified.

KNN selected, with an operating-envelope check



1

Fewest costly errors

over-predicts by more than 5 MW on the fewest hours, and is best at the extremes

2

Most accurate

statistically significant lead on unseen hours

3

Explainable

each prediction is a weighted average of six real past hours

4

Flags what it cannot know

inputs outside the training range or with few similar past hours are marked unreliable

Runner-up: the MLP, least sensitive to noisy inputs. SVR not recommended.

Five unseen cases: sensible values, risky ones flagged

Case	AT (°C)	V	AP (mbar)	RH (%)	KNN (MW)	90% band (MW)	Status
Hot summer afternoon	33	71	1008	45	433.9	428.7 to 439.1	valid
Cold winter night	6	40	1020	85	487.9	482.7 to 493.1	valid
Mild day, humid	20	52	1013	90	453.3	448.1 to 458.5	valid
Mild day, dry	20	52	1013	45	452.6	447.4 to 457.8	**caution: few similar past hours**
Heatwave, 41 °C	41	74	1006	30	429.3	not valid here	**flagged: outside training range**

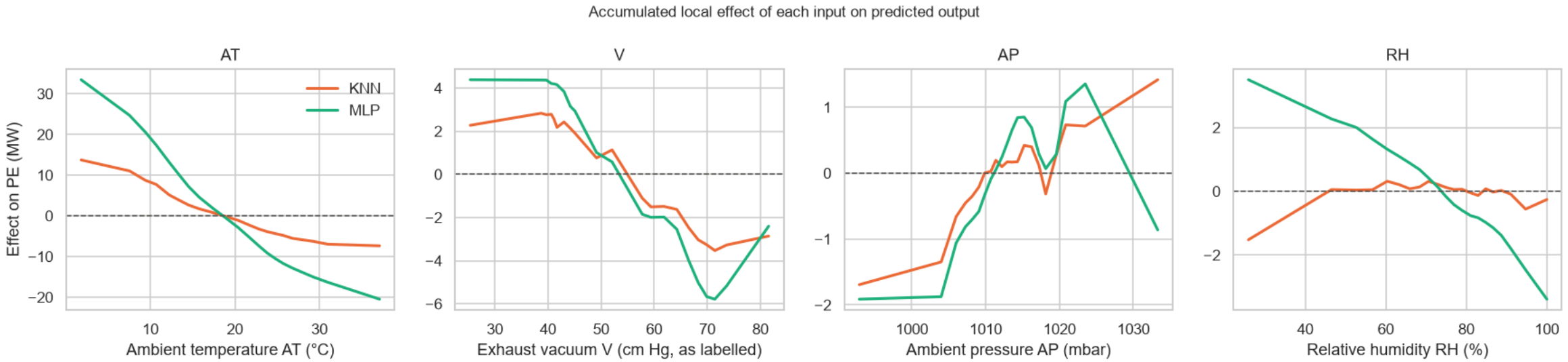
What the cases show

The cold night gives 54 MW more than the hot afternoon, the seasonal swing seen in the data. The heatwave is flagged; there the SVR would predict 449 MW, more than at 33 °C.

A limitation we found

KNN rates the dry day 0.7 MW below the humid day; the data says the reverse. Its neighbours are warmer hours. The density check marks the case "caution".

What the models learned agrees with thermodynamics



−54 MW

AT: across the range; matches the 52 MW seen in the data

−10 MW

V: mostly between 15 and 30 °C; costs 2 to 4 MW per 10 units in spring

−7 MW

RH: dry to humid; confirms the sign reversal

≈ 3 MW

AP: small, no consistent direction, as in the data

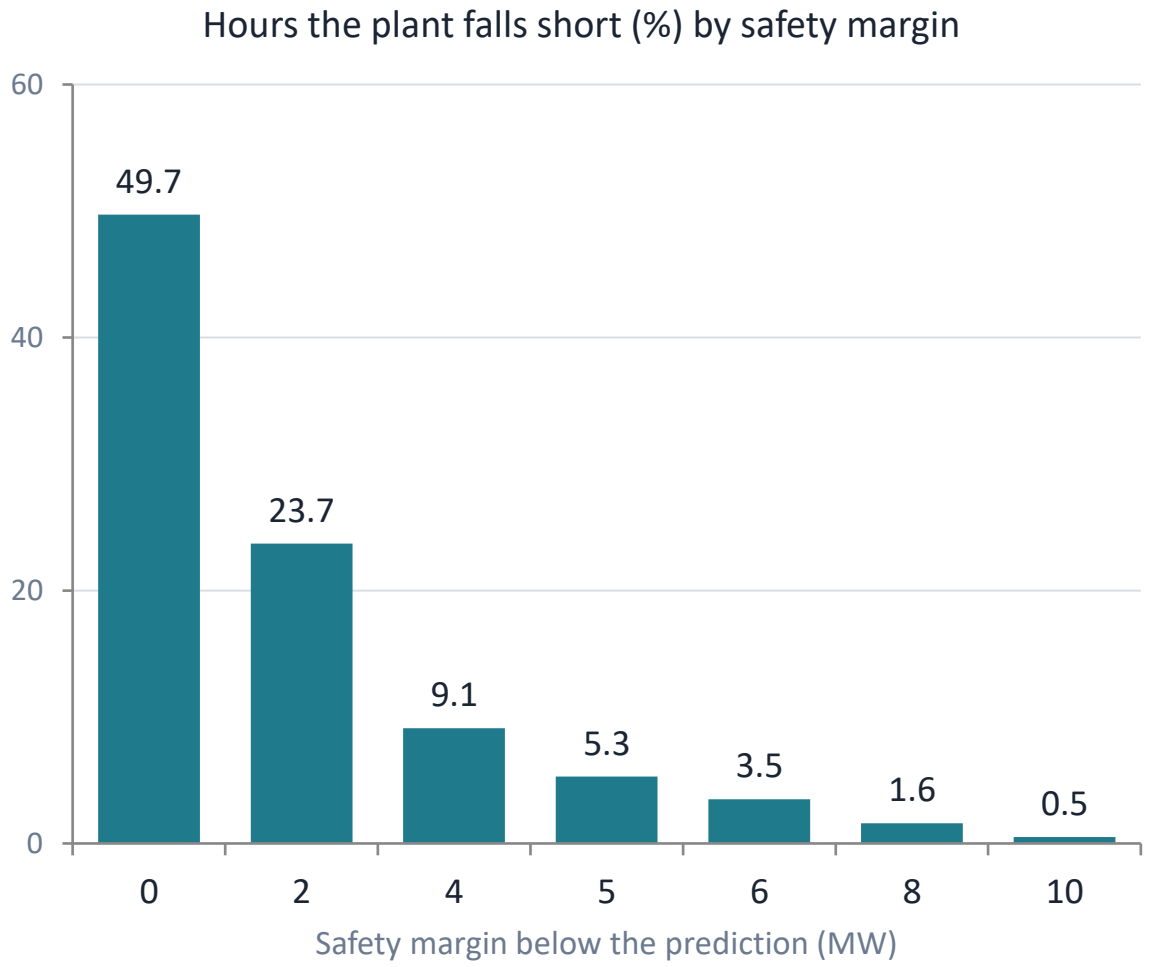
The model supports two plant decisions

Performance alarm

Threshold 5.2 MW: on 95% of normal hours the shortfall is smaller. A 15 MW drop on a cold night raises the alarm; 552 logged hours exceed the threshold.

Safety margin for the grid

Committing the bare prediction falls short on half of all hours. If a missing MWh costs four times its sale price, commit 2.4 MW below the prediction.



Limitations and recommendations

Limitations

- One plant at full load; no transfer to part load or other plants without new data
- No timestamps: no chronological validation, no view of drift over six years
- Trips and maintenance are not recorded, so the largest errors cannot be explained
- V's unit is wrong in the source, so it cannot be read as a true pressure

Recommendations

1. Deploy KNN with both checks for hourly performance monitoring
2. Commit to the grid with a cost-based margin, recomputed after each retraining
3. Use the model without V (AT, AP, RH) for day-ahead commitments: under 1 MW extra error
4. Clean the condenser before the warm season; weigh evaporative inlet cooling against the humidity it adds
5. Log timestamps, trips and maintenance, and calibrate the RH sensor

A validated model the plant can act on

Thank you. Questions?

3.45 MW

test RMSE, MAPE 0.540%

0.959

R^2, against 0.912 in the reference

−54 MW

temperature effect, as physics predicts